Shubhankar Tiwari

Software Engineer | Backend Systems & Applied AI

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Professional Summary

Software Engineer with **4+ years of experience** building **Java/Spring Boot microservices**, **REST APIs**, and **OpenShift**-based backend systems for enterprise fintech. Strong in **CI/CD**, release validation, production incident triage, and applied AI work on **LLM reliability** and quality-gating.

Experience

Apr 2022 – Present **Software Engineer I A**, *Bank of America*, Hyderabad, India

- Develop and maintain client-facing fintech microservices using **Java, Spring Boot, REST APIs, SQL** on **OpenShift**.
- Own backend service delivery: design, implementation, **CI/CD integration**, deployment validation, and production stabilization.
- Lead **production incident triage** — deployment failures, SSL misconfigurations, database connectivity, service dependency conflicts; reduced recurring escalations.
- Perform **cross-functional manual regression testing** during release cycles — validating end-to-end system behaviour across service boundaries before production promotion.
- Improve release stability through proactive validation, pipeline debugging, load-test support, and root cause analysis.
- Contribute to **OpenShift onboarding** and deployment readiness, strengthening infrastructure resilience.

Aug 2021 – Mar 2022 **Product Development Trainee**, *HighRadius Technologies*

Built full-stack modules for **Order-to-Cash and Treasury Systems** using **Java, JavaScript, Python**. Contributed to Agile sprint delivery focusing on backend logic, API integration, and production readiness.

Projects

Jan 2026 – Present **LLM Inference-Time Evaluation**, *Python, Pydantic, Information Theory, FastAPI, Qwen 2.5-7B*

- Built a **probabilistic evaluation system** and designed **3-tier cultural dose-response experiments** to measure inference-time behavioral sensitivity.
- Developed a **9-category guardrail classifier** with severity scaling and adaptive reliability fallbacks based on token entropy.
- Implemented **audit-grade traceability** using Pydantic schemas, SHA-256 artifact fingerprinting, and behavioral snapshots validated by a 55+ file test suite.

Sep – Dec 2025 **Conservative Auto-Regeneration Policy for AI-Generated Financial Narratives**, *Python, ROUGE-L, Sentence Embeddings, Bootstrap Calibration*

Designed a **production quality-gating pipeline** for regulated AI outputs.

- Implemented a **Conservative AND policy** that triggers auto-regeneration only upon dual-metric failure (embedding similarity AND ROUGE-L).
- Calibrated decision thresholds via a **bootstrap percentile sweep** across labeled validation sets to optimize the precision-recall trade-off.
- Built a **priority-scored human review queue** ranked by combined metric deficits and integrated PII sanitization for internal audit compliance.
- Established **audit-grade provenance tracking** recording config snapshots and metadata for full decision reconstruction.

Jun – Aug 2020 **Song Recommender System**, *Python, K-Means, VADER, Scikit-learn*

Co-built a workout playlist recommender using K-Means clustering on lyric sentiment and BPM.

- Identified **Neutrality scores** as the feature with the highest positive covariance with BPM, using it as the primary clustering anchor.
- Optimized K-Means to K=4 clusters via the **WCSS elbow method** to map songs to specific workout intensity tiers (warm-up to aggressive).

Education

2018 – 2022 **B.Tech, Computer Science and Engineering**, *SRM Institute of Science and Technology*, **94.4%**

Certifications & Recognition

Kaggle **Notebooks Expert** — Personal Best #707; 10 bronze medals; 34 notebooks, 11 datasets, 3 models; ML, Deep Learning & Computer Vision certified (2025)

2022 REST API — Etrain

Technical Skills

Backend Java, Spring Boot, Microservices, REST APIs, SQL, OpenShift, Jenkins, CI/CD, Distributed Systems, **Pydantic, FastAPI**

AI/ML Python, PyTorch, HuggingFace Transformers, Scikit-learn, VADER, Sentence Embeddings, ROUGE-L, LLM Evaluation, **Information Theory, KL Divergence**

Practices Production Incident Triage, Regression Testing, Load Testing, Root Cause Analysis, Agile, Release Validation, **Audit-Grade Provenance**

Languages

English (Professional), Hindi (Native)